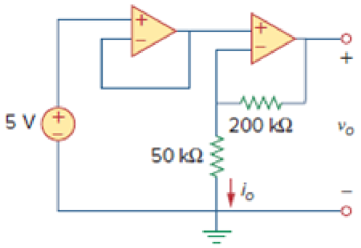
Chapter 5, Problem 9PP

Problem

Determine *vo* and *io* in the op amp circuit in Fig. 5.30.

Figure. 5.30:



Step-by-step solution

Step 1 of 4

Refer to Figure 5.30 in the text book.

Draw the modified circuit with labeled node voltages.

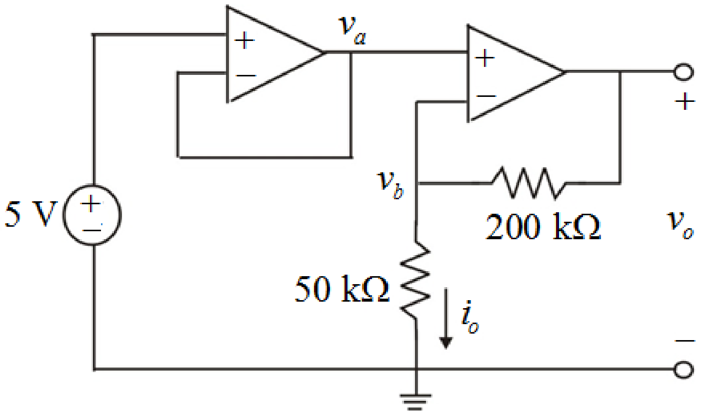


Figure 1

Step 2 of 4

Consider the left-side op-amp:

For an ideal op-amp, the voltages at inverting and non-inverting terminals are same.

$$v_+ = v_- = 5\text{ V}$$

From Figure 1, the value of voltage v_a is,

$$v_a = v_-$$

Substitute 5 V for v_- in the equation.

$$v_a = 5\text{ V}$$

Step 3 of 4

Consider the right-side op-amp:

For an ideal amplifier, the voltages at inverting and non-inverting terminals are same.

$$v_a = v_b = 5\,\text{V}$$

Use voltage division rule, to determine the value of output voltage.

$$v_b = \left(\frac{50\times10^3}{\left(50\times10^3\right)+\left(200\times10^3\right)} \right) v_o$$

Substitute 5 V for v_b in the equation.

$$5 = \left(\frac{50\times10^3}{\left(50\times10^3\right)+\left(200\times10^3\right)} \right) v_o$$

$$\begin{aligned} v_o &= (5)\left(\frac{\left(50\times10^3\right)+\left(200\times10^3\right)}{50\times10^3} \right) \\ &= (5)(5) \\ &= 25\text{ V} \end{aligned}$$

Thus, the value of output voltage, v_o is $\boxed{25\text{ V}}$.

Step 4 of 4

Determine the value of current, i_o.

$$\begin{aligned} i_o &= \frac{v_b}{50\times10^3} \\ &= \frac{5}{50\times10^3} \\ &= 0.1\times10^{-3} \\ &= 100\ \mu\text{A} \end{aligned}$$

Thus, the value of current, i_o is $\boxed{100\ \mu\text{A}}$.